

Development of the Orion Life-Support Integration Facility (OLIF)

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Testing the life support hardware of a vehicle that is going to take humans beyond low-Earth orbit (LEO) in conditions similar to space is crucial. The Orion Life-Support Integration Facility (OLIF) at NASA Johnson Space Center (JSC) was designed and built to test the Orion vehicle's hardware and software as integrated systems to provide a complete Environmental Control and Life Support System (ECLSS) system-level qualification. The existing 11 Foot human rated vacuum chamber has been adapted to accommodate and integrate various qualification and flight like components of the Orion vehicle's Air Revitalization System (ARS), Pressure Control System (PCS), Active Thermal Control System (ATCS) and the Orion Crew Survival System Suits (OCSS). The ultimate goal was to create an analog testbed that could safely support up to four test subjects in open "shirt-sleeve" or closed suit loop configurations and simulate Orion Cabin conditions. This integrated hardware/software ARS and PCS will help identify any technical issues that should be addressed prior to the Artemis-2 mission. This paper will discuss the history of Orion ECLSS hardware development testing in the 11 Foot Chamber, the challenge of integrating flight hardware and software control systems, and the capabilities that make it a unique, world class facility for NASA. It will provide an overview of past and future testing, and the lessons learned along the way.

Acronyms and Nomenclature

ARS	=	Air Revitalization System
ATCS	=	Active Thermal Control System
ATCO	=	Ambient Temperature Catalytic Oxidizer
AVTC	=	ARS Ventilation and Temp Control
CAD	=	Computer-aided Design
CFC	=	Coolant Flow Control Assembly
CHC	=	Carbon Dioxide and Humidity Control Assembly
CO ₂	=	Carbon Dioxide
CPE	=	Cabin Pressure Equalization
CTSD	=	Crew and Thermal Systems Division

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CM	=	Crew Module
DMC	=	Digital Motor Controller
DSC	=	Dual Suit Controller
ECLSS	=	Environmental Control and Life Support System
ECS	=	Environmental Control System
EMI	=	Electromagnetic interference
EMS	=	Emergency Management System
EMU	=	Extravehicular Mobility Unit
ER	=	Emergency Repress
EVA	=	Extravehicular Activity
FSS	=	Fire Suppression System
FSW	=	Flight Software
GN ₂	=	Gaseous nitrogen
GO ₂	=	Gaseous oxygen
<i>gpm</i>	=	Gallons per minute
HITL	=	Human-in-the-Loop
HEPA	=	High efficiency particulate air
ISS	=	International Space Station
IVA	=	Intravehicular Activity
JSC	=	Johnson Space Center
kPa	=	Kilopascal
LCG	=	Liquid Cooling Garment
LEO	=	Low-Earth orbit
LWC1	=	Liquid Water Cooling Assembly
NASA	=	National Aeronautics and Space Administration
NPCC	=	Nitrogen Pressure Control
OCSS	=	Orion Crew Survival System
OLIF	=	Orion Life-Support Integration Facility
OPCC	=	Oxygen Pressure Control
OR2A	=	Oxygen Regulator Assembly
PBD	=	Performance Based Design
PCS	=	Pressure Control System
PLC	=	Programmable Logic Controller
PLSS	=	Portable Life Support System
<i>psia</i>	=	Pounds per square inch absolute
<i>psid</i>	=	Pounds per square inch differential
SCU	=	Service and Cooling Umbilical
SLVA	=	Suit Loop Valve Isolation Assembly
<i>sq. ft.</i>	=	Square foot
TCC	=	Trace Contaminant Control
UIF	=	Umbilical Interface Fixture
UIP	=	Umbilical Interface Panel
xEMU	=	Exploration Extravehicular Mobility Unit
xEVAS	=	Exploration Extravehicular Activity Services

I. Introduction

SINCE the earliest days of space exploration, maintaining control of the environment to which humans are exposed has been critical for crew comfort and safety. Early Mercury missions to low-Earth orbit (LEO) used simple, open-loop environmental control systems that maintained a reduced-pressure, pure oxygen cabin atmosphere, nominally 4.3–5.5 psia (29.64–37.92 kPa) [1], [2]. Crew suits employed open-loop designs in which carbon dioxide (CO₂) was purged overboard rather than actively scrubbed. Temperature control was largely achieved through regulation of makeup oxygen flow [1].

As mission objectives expanded, so did environmental control system requirements. The Gemini and Apollo programs, constrained by mass and consumables for lunar missions, transitioned to closed-loop designs using lithium hydroxide canisters for CO₂ removal, which reduced consumable consumption. These programs, however, continued to operate with pure oxygen cabins at approximately 5.5 psia (37.92 kPa) due to simplicity and mass efficiency [1], [3].

Operational and safety risks associated with low-pressure, oxygen-rich environments became increasingly evident, particularly following the Apollo 1 fire, which highlighted the hazard of high flammability during ground and ascent operations [4]. Subsequent programs adopted mixed-gas cabins at higher pressure to reduce fire risk while maintaining acceptable physiological performance [1].

From the Space Shuttle through the International Space Station (ISS), crewed spacecraft have operated near sea-level pressure (~14.7 psia) with oxygen concentrations of ~21%, closely replicating terrestrial conditions [5]. This evolution required precise control of oxygen partial pressure, active nitrogen management, and continuous removal of metabolic byproducts without reliance on consumable venting. Regenerative air revitalization systems were developed to support long-duration missions while minimizing mass and logistical burden [1], [5].

For deep-space exploration beyond low Earth orbit, integrated ground testing of Environmental Control and Life Support Systems (ECLSS) is increasingly critical. The Orion program, designed to carry crews for extended missions to the Moon and Mars, requires system-level verification of hardware, software, and crew interfaces. To meet this need, NASA developed the Orion Life-Support Integration Facility (OLIF) at the Johnson Space Center (JSC). OLIF provides a human-rated, vacuum-capable test environment for evaluating Orion ECLSS components as an integrated system, enabling early identification of interactions, performance limitations, and operational risks prior to flight. This paper describes the development, capabilities, and testing conducted in OLIF to support Orion ECLSS maturation.

II. Brief History of the JSC CTSD 11 Foot Chamber

The Johnson Space Center Crew and Thermal Systems Division (CTSD) 11 Foot Chamber is a vacuum chamber, 11 feet in diameter. As shown in Figure 1, there are three airlock compartments arranged in series: the Outer Lock, Inner Lock, and Cabin. The chamber is located in Building 7 of the JSC campus in Houston, Texas. The inner-most airlock, dubbed the “Cabin” in reference to the Apollo Lunar Module Cabin, was originally outfitted in the 1960s to conduct crewed testing of the Apollo Lunar Module Environmental Control System (ECS) interfaces.

During the 1980s, the first inner lock served as a human-in-the-loop test facility for the development and certification of the Space Shuttle and International Space Station Extravehicular Mobility Unit (EMU). It was equipped with dual vacuum-rated treadmills and weight-relief systems to support vacuum testing of the EMU through complete Extravehicular Activity (EVA) metabolic profiles.

Within the last few years, the facility has also been upgraded to support testing of the next-generation spacesuits including the ones being designed and built by commercial companies through NASA’s Exploration Extravehicular Activity Services (xEVAS) program. These new suits are developed to support space exploration beyond LEO to the Moon, Mars, and beyond.

During the 2010s, the 11 Foot Chamber was chosen to also serve as the testbed for a series of Pressure Integrated Suit Tests (PIST) to characterize the development of Orion Environmental Control and Life Support System components.

III. OLIF Test Bed

The Orion Program established the Orion Life-Support Integration Facility to provide integrated performance testing of the Orion ECLSS in support of Artemis-2. To accomplish this, personnel from NASA's CTSD and Lockheed Martin's Orion ECLSS teams integrated ECLSS hardware (HW) and flight software (FSW) within the 11 Foot Chamber.

OLIF's primary objective is system-level hardware verification, encompassing 26 Orion ECLSS and 4 Orion Crew Survival System (OCSS) suit requirement

verifications. While all Artemis-2 hardware underwent extensive component-level testing, including thermal, electromagnetic interference (EMI), and life-cycle evaluations, OLIF is the only facility where these components operate together as a fully integrated system. Key subsystems under evaluation include the Orion ECLSS Pressure Control System (PCS), Air Revitalization System (ARS), and OCSS suit.

As part of the integrated testing effort, the OLIF team developed a comprehensive Test Plan defining three primary test cases: Cabin Pressure Control, Feed the Leak, and Suited Test.

- Cabin Pressure Control assesses the PCS's ability to maintain target cabin pressure and oxygen concentration under shirt-sleeve conditions.
- Feed the Leak evaluates contingency operations by simulating a leak equivalent to a 0.25 in. (6.35 mm) hole in the cabin.
- Suited Test scenarios examine human-in-the-loop (HITL) operations across multiple configurations.

Because these systems are being operated together for the first time with flight software, all initial test points are performed unmanned, without humans in the loop.

The Test Plan also specifies hardware allocations, test environments, a simplified mechanical layout of the 11 Foot Chamber, and configurations for each scenario. During testing, the chamber operates from a vacuum of 0.2 Torr (26.6 Pa) to ambient pressure while maintaining defined setpoints. Materials must be compatible with up to 40% oxygen in the chamber and up to 100% oxygen in the suit loop. Sensor instrumentation monitors hardware health and ensures safe conditions for test personnel.

By enabling integrated system operation for the first time, OLIF provides critical verification of ECLSS performance and early identification of potential system-level issues, thereby directly supporting Artemis-2 mission readiness.

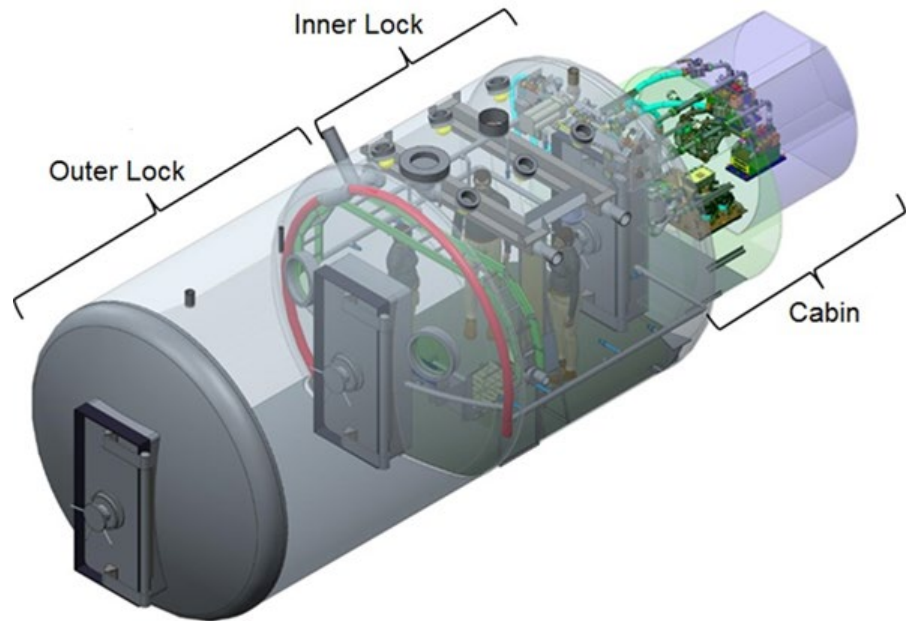


Figure 1. 11 Foot Chamber CAD model with the three airlock compartments.

IV. OLIF Implementation

The OLIF project was initiated in June 2019, with testing originally scheduled for spring 2021. The design and buildup of OLIF posed several technical challenges, primarily due to spatial constraints and the multiuse of the 11 Foot Chamber. The chamber serves a critical role in the current generation EMU certification cycle and is planned for use with next-generation exploration EVA suits. As a result, OLIF design had to maintain existing chamber capabilities while minimizing reconfiguration effort between tests.

The chamber's physical constraints required careful management of reserved spaces and previously installed hardware, including the EMU treadmill, treadmill support stand, handrails, and the EMU Service and Cooling Umbilical (SCU) panel. Additionally, provisions were required for the installation of the Exploration Extravehicular Mobility Unit (xEMU) donning fixture, related hardware, and xEMU services panel. To address these constraints,

System testing commenced in an unmanned configuration to verify proper operation of both Orion hardware and facility systems. Prior to human in the loop testing, emergency systems were confirmed operational, and the purity of supply gases was verified to meet safety requirements. This approach ensured safe and reliable conditions for subsequent human testing within the 11 Foot Chamber.

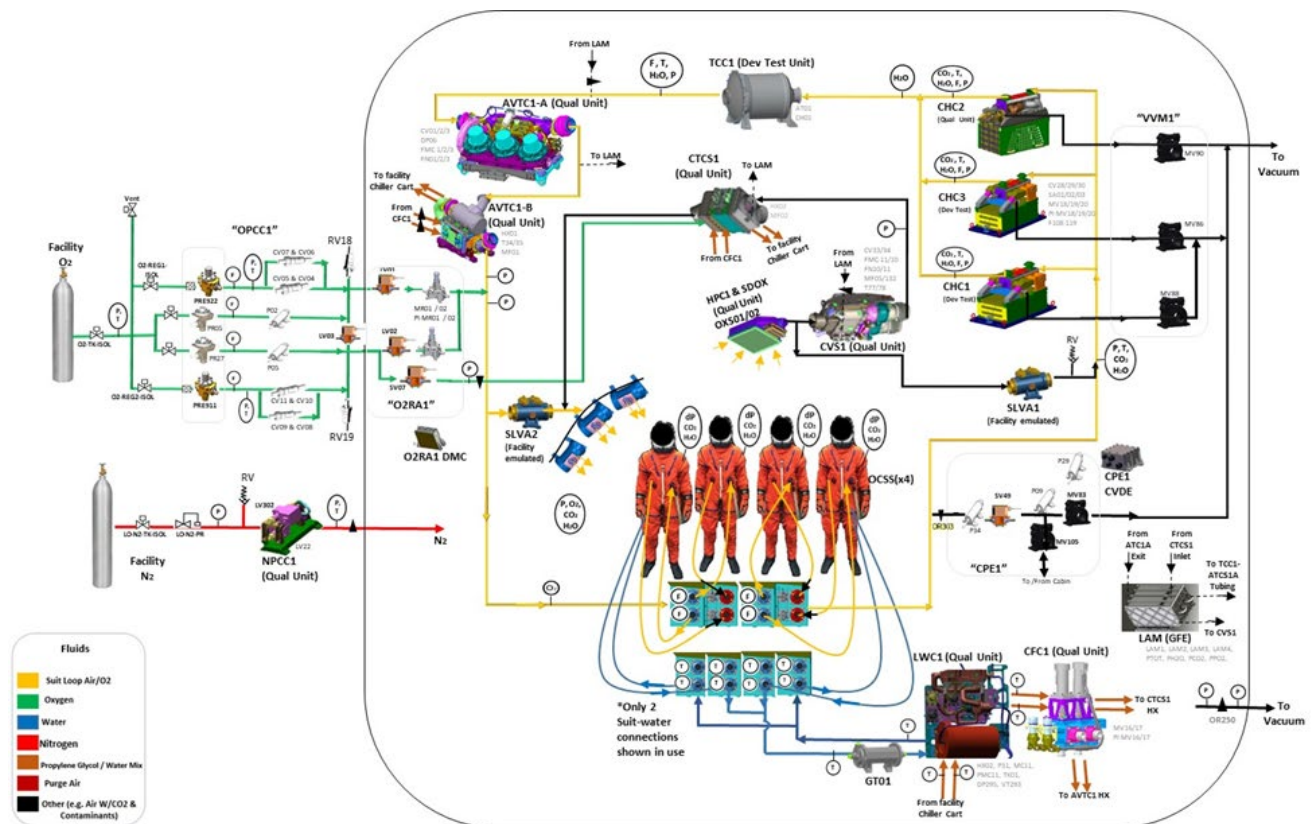


Figure 2. OLIF schematic.

A. Air Revitalization System

The existing ARS, which has evolved through the years, was focused only on suited testing. It was designed to incorporate the available development hardware and utilized commercial off the shelf components to simulate suit loop hardware that was not available. For the OLIF, development ARS hardware such as the ARS Ventilation and Temp Control (AVTC) and Cabin Pressure Equalization (CPE) subassemblies were replaced with Artemis-2 qualification hardware used to qualify the Artemis-2 flight hardware subassemblies. Due to the limited availability of certain flight design hardware several development components continue to be used for OLIF such as the Trace Contaminant Control (TCC) subassembly, which uses a combination of activated charcoal and Ambient Temperature Catalytic Oxidizer (ATCO) to control odors and trace contaminants. The TCC is sized and designed to remove a four person load of human metabolic trace contaminants as well as those generated by the ARS components. In order to mimic the Artemis-2 ARS, the OLIF ARS was designed to incorporate three Carbon Dioxide and Humidity Control (CHC) units. Because of delays in testing and certification of the flight design qualification CHC, initial testing will utilize the two development CHCs.

The CHC is a regenerative technology that adsorbs metabolic carbon dioxide (CO₂) and moisture (H₂O) from the ARS and is regenerated by exposure to vacuum which causes the CO₂ and H₂O to desorb. Vacuum pressure is crucial to the effectiveness of the CHC performance. Degradation of CHC performance results in excess consumable usage and potential inability to control CO₂ levels. The OLIF CHC vacuum source is a system consisting of two vacuum boosters in series backed by a rotary screw vacuum pump. A dry screw vacuum pump was chosen for the ability to handle high water loads without the loss of performance seen with extended operation of oil sealed pumps operating at the same conditions. Total vacuum system throughput is 3600 acfm (102 m³/min) and is capable of maintaining a minimum delivered vacuum pressure of 0.5 torr (67 Pa) while pumping a gas load of 1.5 lbm/hr (0.68 kg/hr) CO₂ and 4.0 lbm/hr (1.81 kg/hr) H₂O.

Sensors within the ARS provide critical system insight for science and safety. Dewpoint, humidity, and CO₂ sensors are used to monitor real time CHC performance and provide analytical data to quantify metabolic loads produced during crewed testing.

B. Pressure Control System

The OLIF Pressure Control System (PCS) is designed to mimic that of the Artemis-2 PCS, consisting of four major subsystems; the Oxygen Pressure Control (OPCC) subassembly, Oxygen Regulator Assembly (O2RA), Nitrogen Pressure Control (NPCC) subassembly, and Cabin Pressure Equalization (CPE) subassembly. Controlled by Artemis-2 flight software, the four subsystems are used to control the 11 Foot Chamber inner lock pressure and atmosphere.

The OPCC assembly is used to control high pressure oxygen supply to the O2RA. OPCC consists of four pre-set high pressure oxygen regulators, four quad check valves, pressure transducers and a latching solenoid valve. On the Artemis-2 vehicle the pre-set high pressure oxygen regulators are located outside of the pressurized volume and reference external vehicle pressure. In order to achieve the required reference pressure the regulators are installed in a small external vacuum chamber, vacuum is provided by a small scroll pump.

The O2RA is used to control oxygen introduction for suit pressure control, cabin pressure control, for cabin repressurization and for contingency cabin feed the leak operation. O2RA consists of two motor-settable pressure regulators, two latching valves, one solenoid valve, and a Digital Motor Controller (DMC). Suit pressure is controlled by the motor-settable regulators through the DMC and flight software.

The NPCC assembly is used to control gaseous nitrogen introduction into the cabin for total cabin pressure maintenance, for cabin repressurization, or for contingency cabin feed the leak operation. NPCC consists of two solenoid valves. The first can be closed to prevent cabin over-pressurization due to failed upstream components. The second will be opened by FSW command when nitrogen is required by the Crew Module (CM) compartment and closed when the chamber is within the target pressure range.

In OLIF, the NPCC controls nitrogen introduction into the inner lock for chamber pressure control and feed the leak testing. The inlet pressure range of NPCC is 34.4–50 psia (237–344 kPa). Therefore, in OLIF a gaseous nitrogen pressure regulator (facility supplied) is used to reduce the facility nitrogen supply pressure from 2000 psig (13890 kPa).

The CPE subassembly is able to vent suit loop pressure to the chamber volume or overboard to the facility vacuum system. The CPE is used for suit and/or cabin pressure equalization. In OLIF, CPE is responsible for relieving the suit loop pressure during depress and returning the suit loop to 0 psid (0 Pa) with the chamber when testing is complete. The CPE also allows for a purge capability to induce makeup oxygen from the primary oxygen source and regulator if desired. The CPE is designed with two motor valves, one solenoid valve, and three pressure sensors for monitoring the pressure and releasing either the cabin or the suit loop pressure when required. During cabin depressurization test

cases the CPE, controlled by flight software, reduced the pressure of the 11 Foot Chamber inner lock while a technician manually controls the inner-most lock of the chamber to maintain a small differential pressure across the chamber hatch to ensure a good seal to maintain the integrity of the test atmosphere.

C. Cabin ARS

The Crew Module cabin loop provides the necessary airflow for cabin air circulation, for particulate removal through the HEPA filter, and across the Cabin Heat Exchanger to reject heat in order to provide cabin temperature control. When the crew is unsuited, the Suit Loop Valve Isolation Assemblies (SLVAs) are open, which allows the Cabin and ARS loops to share a common inlet through the HEPA filter and outlet through cabin diffusers.

D. Flight Software/Controls

The OLIF test configuration utilizes a high-fidelity, integrated software simulation of the Orion flight vehicle to execute flight software and emulate vehicle-level dynamics, including subsystem behavior, sensor feedback, and actuator responses. This simulation environment enabled closed-loop operation between the flight software and the OLIF hardware, providing command generation, telemetry acquisition, and real-time monitoring to support functional performance assessment under representative mission conditions. The software was hosted on a dedicated test rack located outside the vacuum chamber and interfaced with the OLIF hardware via extended electrical harnessing. These harnesses were routed through chamber penetration plates and incorporated vacuum-rated connectors to preserve chamber integrity while supporting power distribution, command and data exchange, and sensor signal routing between the external software rack and the hardware under test.

E. Suit Integration

The overall design of the pressure suit is unchanged from previous pressure integrated suit test series. The Orion Crew Survival System suit will be used for the Orion Artemis-2 mission, and therefore it will be part of the OLIF test. As seen in Figure 3, the OCSS suits are “soft suits” that can operate in a closed loop system. These are full pressure suit assemblies that consist of a coverall assembly, gloves, helmet, and boots. For OLIF, the suits being used will be flight-like class II units with the latest design features not utilized before (feed port, urine collection, manned Dual Suit Controller ops). Test subjects are semi-suspended in the 11 Foot Chamber inner lock due to the duration of suited tests, upwards of 10 hours, and limited OCSS suit mobility when pressurized. A spreader bar and harness attached to existing weight relief structures allow suited test subjects a small degree of freedom of movement to stand or the ability to sit and offload the weight of the suit.

The OCSS suits do not contain a Primary Life Support System (PLSS), therefore their system relies upon the OLIF ECLSS and safety features integrated into the ARS loop, including pressure control, oxygen delivery control, and CO₂ control. The Orion Program life support Intravehicular Activity (IVA) umbilicals provide closed-loop gas for suit pressurization and breathing, closed-loop cooling fluid and flow control, and wiring for hardline communications. These umbilicals are designed to be used for IVA that include pressurized and unpressurized environments and conditions that include launch, landing, and all dynamic flight phases for the Orion vehicles. The umbilicals connect to the OLIF ECLSS via an Umbilical Interface Panel (UIP) located in the 11 Foot Chamber inner lock. The UIP is outfitted with Facility delta pressure and flow sensors and provides a manifold for the suit umbilicals to integrate to the ARS Loop. Umbilical Interface Fixtures (UIF) provide the interface for the umbilicals to connect the ARS Loop. The UIF provides connectors for gas flow in and flow out.



Figure 3. Four test subjects in OCSS suits in the 11 Foot Chamber.

F. Active Thermal Control System

The Active Thermal Control System (ATCS) implemented for the OLIF test configuration consisted of a limited subset of the vehicle thermal control architecture necessary to provide active thermal regulation for the cabin environment and suited crew interfaces. While the flight vehicle ATCS includes additional components and subsystems, only the elements required to meet OLIF thermal control objectives were incorporated. Specifically, the

Coolant Flow Control Assembly (CFC1) and the Liquid Water Cooling (LWC1) assembly were implemented to enable active control of air and liquid cooling loops. The CFC1 distributes and regulates coolant flow between the Air Revitalization System and cabin heat exchangers, allowing control of cabin and suit loop air temperatures. The assembly consists of two motorized mixing valves, one per coolant loop, and associated rotor position sensors. The LWC1 provides liquid water cooling to the suited crew through suit umbilicals that supply water to the Liquid Cooling Garments (LCG). The LWC1 includes an accumulator tank, pump and motor controller, heat exchanger, pressure sensing, and flow and differential pressure measurement instrumentation. Together, these components provided the necessary active thermal control capability to support OLIF testing while maintaining a representative yet constrained implementation of the vehicle ATCS.

G. Emergency Systems

Since the 11 Foot Chamber is a HITL facility, various emergency systems are in place for safety of the test subjects and test teams. The Emergency Management System (EMS) implements predetermined actions in the case of prescribed emergencies that may be encountered during testing. The system has a primary and backup Programmable Logic Controller (PLC). This system covers an emergency repress (ER), which can repress the inner lock in less than one minute. This would be used in the case of an incapacitated test subject that needs to be rapidly evacuated from the inner lock. In addition, the EMS can also kill power within the chamber in case of a fire or smoke in the facility.

Inside and outside of the chamber, there are multiple cameras for observing the test subjects. In case of a power kill inside of the facility, the external cameras are kept on to continue watching the test subjects during the emergency. At all times, the external test team has communications with the test subjects within the chamber. If there are loss of comms, rescue technicians can communicate to the subjects using preset cue cards through the chamber view ports. All of the videos, communications, and facility data are recorded. The existing 11 Foot Chamber fire suppression system has been in place for a long time. Due to the target chamber atmosphere during OLIF Cabin test days, CTSD System Test Branch facility engineers were concerned that the system would not meet national consensus codes and standards, as well as NASA/JSC requirements for fire protection. As a result, the Systems Test Branch included the 11 Foot Chamber in to a Performance Based Design (PBD) of the JSC Building 7 crewed chambers to identify deficiencies related to applicable code and standards and to derive a PBD that ensured an equivalent level of safety during test operations. The study concluded that the chamber was designated an NFPA 99B, Class E, oxygen enriched atmosphere based on the OLIF requirements, which per code would necessitate a water spray density of 7.5 gpm/sq.ft. (0.005 cubic meters per second per square meter). The PBD developed out of the chamber study established a set of alternative code equivalent requirements in accordance with NASA STD 8719.11 and established an upper chamber atmosphere oxygen concentration limit of 40%. The PBD further established a minimum 2.0 gallons per minute per square foot (0.001 cubic meters per second per square meter) water density as sufficient. To comply with the PBD in the 11 Foot Chamber, the deluge nozzles were replaced with new 2.9K nozzles listed for fire suppression. The 11 Foot Chamber Outer Lock supply piping was connected to a new 8 inch pipe that was installed as part of another FSS upgrade in Building 7.

V. Conclusion

OLIF represents a critical capability for advancing integrated ECLSS verification in support of human exploration beyond low-Earth orbit. By adapting the JSC's 11 Foot Chamber into a flight-representative, software-integrated testbed, OLIF enables system-level evaluation of Orion ECLSS hardware, flight software, and crew interfaces operating together under relevant environmental conditions. This integrated approach provides insight that cannot be achieved through component- or subsystem-level testing alone.

OLIF testing has demonstrated the feasibility and value of operating Air Revitalization, Pressure Control, Active Thermal Control, and suit systems as a closed-loop architecture using flight software control. The facility enables early identification of system interactions, operational sensitivities, and performance limitations, thereby reducing technical risk prior to crewed flight. The incremental buildup strategy, use of unmanned testing prior to human-in-the-loop operations, and incorporation of facility safety systems further establish a robust framework for conducting complex life-support verification in a controlled and safe manner.

Beyond its immediate role in supporting Artemis-2, OLIF provides a flexible platform for future Orion missions and other exploration programs requiring integrated life-support testing. The lessons learned from facility implementation, hardware integration, and test execution inform best practices for future ground-based system qualification. As mission durations increase and life-support architectures evolve, facilities such as OLIF will remain

essential for validating system performance, ensuring crew safety, and enabling sustainable human presence in deep space.

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